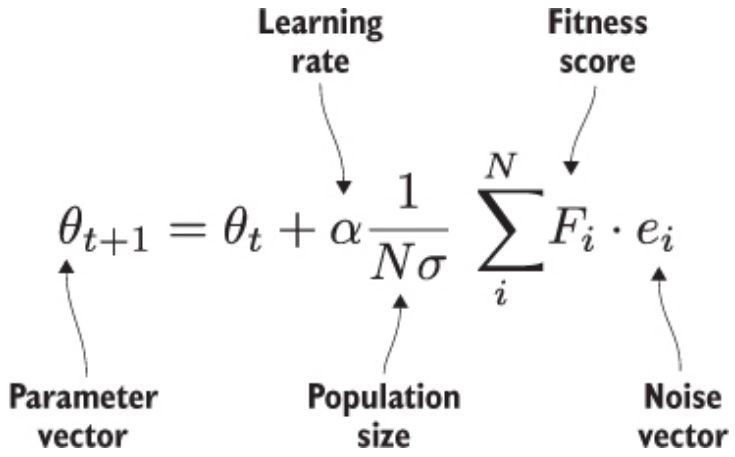



This evolutionary strategy algorithm is significantly simpler than the genetic algorithm we implemented earlier since there is no mating step. We only perform mutation, and the recombination step does not involve swapping pieces from different parents but is just a simple weighted summation which is very easy to implement and computationally fast. As we'll see, this approach is also easier to parallelize.

### 6.5.2. Parallel vs. serial processing

When we used a genetic algorithm to train agents to play CartPole, we had to sequentially iterate over each agent and let each agent play CartPole until it lost, in order to determine the fittest agent in each generation before we started the next run. If the agent takes 30 seconds to run through the environment, and we are determining the fitness for 10 agents, this will take 5 minutes. This is known as running a program *serially* (figure 6.12).

**Figure 6.12. Determining the fitness of an agent is often the slowest step in a training loop and requires that we run the agent through the environment (possibly many times). If we are doing this on a single computer, we will be doing this in serial—we have to wait for one to finish running through the environment before we can start determining the fitness of the second agent. The time it takes to run this algorithm is a function of the number of agents and the time it takes to run through the environment for a single agent.**